

Cross-Environment Human Activity Recognition from WiFi CSI: A Controlled Study of Adversarial and Contrastive Training Objectives

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Abstract—WiFi Channel State Information (CSI) enables device-free human activity recognition, but models trained in one environment degrade when deployed in another, making cross-environment generalization the main obstacle to practical use. We present a controlled study on the SHARP micro-Doppler dataset under a leave-one-environment-out protocol: starting from a faithful reproduction of the reference SHARP model, we train a deep residual network with plain cross-entropy (CE) and ask whether two popular representation-shaping objectives — an adversarial domain classifier trained through a gradient reversal layer (GRL) and a supervised contrastive (SupCon) loss — improve transfer to the unseen environment. All configurations share one backbone, one data pipeline and one evaluation harness; the test set is evaluated once, with validation-selected checkpoints, and differences are assessed with a trace-level paired bootstrap. The deep CE model transfers best, reaching macro-F1 0.804 on the held-out environment against 0.543 for the SHARP-style shallow network in the identical recipe — a 23.5-point accuracy gap resolved by the bootstrap. Neither objective helps: SupCon loses 7.8 accuracy points (bootstrap-resolved), and the GRL loses 15.8 with a $10\times$ larger observed seed spread. Frozen-encoder probes explain why: no domain attribute is linearly readable from the CE features, so the adversary has no target and can only inject gradient noise. Test-time adaptation (T3A, AdaBN) does not close the gap, and an amplitude augmentation acts as a generic regularizer rather than a cross-domain lever. Our results suggest measuring the readability of the nuisance factor before investing in invariance objectives, and provide a reproducible negative baseline for CSI HAR benchmarking.

Index Terms—WiFi sensing, Channel State Information, Human Activity Recognition, Domain Generalization, Contrastive Learning, Adversarial Training.

I. INTRODUCTION

Wireless sensing turns the existing WiFi infrastructure into a device-free perception system: a moving person perturbs the multipath propagation of the radio signal, and these perturbations are recorded in the Channel State Information (CSI) exposed by commodity network cards. Human Activity Recognition (HAR) from CSI requires no wearable, no camera and no cooperation from the subject, which makes it attractive for assisted living, surveillance and smart-home applications [1]. The micro-Doppler signature derived from the CSI — the distribution of reflected power over velocity

and time — is among the most informative representations, and is the input we work with throughout this report.

The obstacle that separates CSI HAR from deployment is not in-distribution accuracy, which is by now high, but *generalization across environments*. A model trained in one room entangles the activity with the room’s static geometry (walls, furniture, transmitter–receiver placement); moved to a new room, the same activity produces a different multipath fingerprint and accuracy drops. The literature has responded with two families of representation-shaping objectives: *adversarial domain-invariance*, where a domain classifier trained through a gradient reversal layer (GRL) [2] pushes the encoder to discard environment-specific cues, and *supervised contrastive* (SupCon) learning [3], which promises a more transferable embedding geometry. Whether these objectives actually help on a given benchmark is usually assumed rather than tested under controlled conditions.

In this work we test exactly that, on the SHARP dataset [4] under a leave-one-environment-out protocol: with backbone, data pipeline, sampler, seed and evaluation harness held identical, does adding a GRL or replacing cross-entropy (CE) with SupCon improve transfer to an unseen environment? Our answer is a controlled negative result, together with a measured explanation of why the adversarial route in particular could not have succeeded on this class of datasets.

Contributions. The key contributions of this paper are:

- A controlled comparison, under one shared experimental apparatus and a single-use test protocol, of CE, CE+GRL and SupCon for cross-environment CSI HAR, anchored to a faithful reproduction of the SHARP reference model. Our deep CE model reaches macro-F1 0.804 on the held-out environment, against 0.543 for the SHARP-style shallow network in the identical recipe.
- Evidence that neither objective improves over plain CE: SupCon loses 7.8 accuracy points (bootstrap-resolved) and the GRL loses 15.8 with a $10\times$ larger observed seed spread than CE.
- An explanation via frozen-encoder diagnostics: no domain attribute is linearly readable from the learned features, so the adversary has no target — a structural property of training splits dominated by a single environment,

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which we find replicated across every loss function and both held-out-environment rotations we probed.

- A statistical protocol suited to small LOEO test sets: trace-level paired bootstrap, explicit separation of test-sampling and seed variance, and declared multiplicity — under which we also show test-time adaptation (T3A, AdaBN) and amplitude augmentation do not close the gap.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the processing pipeline, Section IV the dataset and features, and Section V the learning framework and the evaluation protocol. Section VI presents the experimental results and Section VII offers concluding remarks.

II. RELATED WORK

WiFi CSI human activity recognition. Early CSI HAR pipelines fed hand-crafted statistics of the amplitude/phase time series to shallow classifiers; the field moved decisively to deep models operating on time–frequency representations, of which the micro-Doppler spectrogram is the most common [1]. The reference point for this report is SHARP [4], which extracts a Doppler representation from commodity IEEE 802.11 CSI by removing the phase offsets and reconstructing the channel, and classifies activities with a comparatively shallow, wide convolutional network followed by a decision-fusion step across antennas. SHARP explicitly targets environment- and person-independence, and its own evaluation already shows accuracy degrading on unseen rooms; that cross-environment setting, where the geometry at test time is unseen, is the one we adopt and push further with a strict leave-one-environment-out protocol. We reproduce the SHARP model as our C0 anchor and reuse its antenna decision-fusion for that configuration, so that our comparison is anchored to a published, faithful baseline rather than to an in-house model of unknown strength.

Domain generalization by adversarial invariance. The dominant recipe for making a representation domain-invariant is adversarial: a domain discriminator is trained on the encoder features while a gradient reversal layer (GRL) [2] inverts its gradient into the encoder, so that the encoder is optimized to make the domain *unpredictable* while the task head still classifies. DANN-style training has been transplanted into CSI sensing to remove environment- and subject-specific nuisance factors. The implicit precondition, rarely tested, is that the nuisance factor is actually present and readable in the features; when it is not, the adversary reduces to a gradient of noise. A central empirical point of this report is to *measure* that precondition directly on our encoder, rather than assume it, and to show that the adversary’s cost is paid whether or not the target exists.

Contrastive representation learning. Supervised contrastive learning (SupCon) [3] extends the self-supervised InfoNCE objective [5] by treating all same-label samples as positives, and has been reported to yield representations that

are more robust and transferable than those trained with cross-entropy. The promise is geometric — a class-collapsed, well-separated embedding — but whether that geometry survives a linear (or any) readout, and whether it transfers better than CE across environments, is empirical. We evaluate SupCon under the identical LOEO protocol and probe its representation with linear, nearest-class-mean and kNN readouts to test the transferability claim head-on.

Test-time adaptation. When some unlabeled target data is available at inference, test-time adaptation can partially close the domain gap without retraining: AdaBN [6] re-estimates batch-normalization statistics on the target distribution, and T3A [7] adjusts the classifier prototypes from confident target pseudo-labels. These are complementary to the training-time objectives above, and we include them as pre-registered rows on our CE model to bound how much of the residual gap is recoverable purely at test time under our deployment assumptions.

Positioning. Prior work overwhelmingly reports the cases where a domain-invariance or contrastive objective *helped*; the complementary evidence — a controlled setting in which they do not, with a measured explanation of why — is underreported, and is what this report contributes.

III. PROCESSING PIPELINE

The system maps a raw CSI recording to an activity label through four stages, shared identically by every configuration we compare: (i) a *Doppler front-end* that turns each antenna’s CSI into a micro-Doppler time–velocity map; (ii) a *windowing* stage that segments each recording into fixed-length windows and normalizes them with training-only statistics; (iii) a shared convolutional *encoder* mapping a window to a feature vector; and (iv) a task *head* producing per-window class scores, fused across the four antennas into one prediction per window. The configurations differ *only* in the training objective attached to the encoder (plus, for the adversarial ones, a domain head that exists at training time only): front-end, windowing, encoder architecture, sampler, fusion and evaluation harness are byte-for-byte the same. This single-apparatus design is what makes the comparison a test of the loss family rather than of incidental engineering differences.

IV. SIGNALS AND FEATURES

Dataset. We use the SHARP micro-Doppler dataset [4], collected with identical IEEE 802.11 hardware in *three environments* — a bedroom, a living room and a laboratory — with three subjects performing eight activities, denoted by the dataset’s own letter codes C, E, H, J, L, R, S, W. Five of them form the reference paper’s core set — E (empty room), W (walking), R (running), J (jumping) and L (sitting still) — and the remaining three (C, H, S) come from the extended set, C being sit-down/stand-up. The recordings are organized into seven capture sets over twelve campaigns; each physical recording (a *trace*) is captured on four antennas. The environment structure enables a leave-one-environment-out evaluation; its imbalance — the bedroom dominates the

collection, and the living room appears in a single capture set — turns out to be central to our findings (Section VI).

Pre-processing and features. From each antenna’s CSI we obtain the micro-Doppler representation as in the reference pipeline, yielding a time-velocity map per antenna. We deliberately did not re-engineer the front-end: keeping it fixed isolates the variable under study, the training objective. Because the velocity and time axes carry a physical orientation, velocity flips and time reversal are excluded from augmentation.

Windowing and normalization. Each trace is segmented into fixed-length windows with a fixed hop (a larger hop at evaluation time). Normalization statistics (μ, σ) are computed on the training split only, per rotation, stored in the frozen split file, and reused identically on validation and test.

Splits. Two rules govern every split and are enforced by blocking assertions in the code. First, *we split by trace, never by window*: all windows of a trace, on all four antennas, fall entirely on one side of the train/val/test partition — windows of the same trace on both sides would leak the static channel and inflate every metric. Second, splits are frozen once as JSON files under version control and never edited. The primary rotation holds out the laboratory for test (11 traces); a second rotation holds out the living room (15 traces) and is used for replication. Rare activity-environment cells are pinned into training to guarantee coverage. One consequence is declared up-front: the small validation split does not contain every class (no C, E, S traces in the primary rotation), so validation macro-F1 is a 5-class, within-run selection signal, not comparable to the 8-class test metric — and in Section VI we test empirically whether this selection blindness harmed the affected classes.

V. LEARNING FRAMEWORK

Shared backbone. All configurations except C0 share a deep residual convolutional encoder [8] (≈ 2.8 M parameters, feature dimension $d_{\text{enc}} = 256$), chosen after a throughput gate on the target hardware. C0 uses a faithful reimplement of the shallow, wide SHARP network [4], so the reproduction is architecturally honest; the same network is also dropped into the exact C1 recipe as a backbone ablation (*C1-sharplike*), so that the backbone axis is isolated with everything else held fixed.

Configurations. Tab. 1 summarizes them. C0 is the baseline (the prior approach); C1 is our model; C2–C4 are its objective variants, spanning a 2×2 design over loss family (CE vs. SupCon) and adversary (none vs. GRL):

- **C0** — SHARP reproduction on the paper’s 5-class core and split, with the paper’s antenna decision fusion.
- **C1** — CE with label smoothing on the shared backbone; our model, and the internal control for C2–C4.
- **C2** — C1 plus a capture-set classifier trained through a GRL [2]: $\mathcal{L} = \mathcal{L}_{\text{task}} + \beta \lambda(p) \mathcal{L}_{\text{env}}$ with a fixed ramp $\lambda(p)$.
- **C3** — SupCon [3]: a contrastive phase trains encoder and projection head with a $P \times K$ sampler and two augmented

TABLE 1: Configurations. C0 is the prior-approach baseline, C1 our model, C2–C4 its objective variants. All but C0 share one backbone, pipeline, sampler and harness.

Cfg	Loss	Adversary	Role
C0	CE (SHARP)	—	baseline (prior approach)
C1	CE	—	our model / CE control
C2	CE	GRL	adversarial invariance
C3	SupCon	—	contrastive
C4	SupCon	GRL	closed on evidence

views per sample; a frozen-encoder linear probe then selects and evaluates the checkpoint. A CE fine-tune of the whole network from the SupCon initialization (C3-*ft*) gives the representation a fair shot beyond the frozen probe.

- **C4** — SupCon+GRL. *Closed on evidence, not trained*: the diagnostics of Section VI-D show no readable domain target under either loss family, so C4 could only reproduce “C3 plus noise”. We report this as an evidence-based triage decision rather than spend its budget confirming a foregone null.

Sampler, optimization, reproducibility. The contrastive configuration uses a $P \times K$ sampler drawing K distinct traces from each of P classes per batch, with a minimum reuse offset so a batch never contains near-duplicate windows of one trace; gradient accumulation is forbidden in the contrastive phase, as it would corrupt the in-batch positive/negative structure. A single seed (42) governs initialization, per-epoch sampler reseeding and augmentation; seed-43 replicates of C1 and C2 measure the training-variance floor. Differences within that floor are treated as comparable, not as improvements.

Evaluation harness and the single-use test rule. One shared harness maps a checkpoint to per-set metric files; C0’s decision fusion is a thin wrapper over the same harness. Checkpoint selection uses validation only, and the *test set is evaluated exactly once*, at the end, in a single session over a frozen, pre-registered list of 16 evaluation rows — every test access is logged by the harness before any test data is read. No intermediate test evaluation and no evaluate-then-decide selection is possible under this protocol.

Test-time adaptation. On our model only, and pre-registered before the session, we evaluate AdaBN [6] (full-pass re-estimation of batch-norm statistics on the unlabeled test set), T3A [7] (a batch variant adjusting the classifier prototypes from confident test pseudo-labels), and their combination, with all hyperparameters fixed a priori. These bound how much of the residual gap is recoverable at inference time. We also evaluate a pre-registered amplitude-scaling augmentation arm (*C1-aug*) on both rotations, with a declared discriminator between a cross-domain effect and generic regularization (Section VI-E).

VI. RESULTS

We first state the statistical protocol, then present the main test table, the comparison against the SHARP baseline, the effect of the two training objectives, the diagnostics that explain

TABLE 2: Main results. Validation macro-F1 (checkpoint selection), test macro-F1 and accuracy on the held-out environment, and the paired accuracy difference vs. our model C1 with its 95% trace-level bootstrap CI (* = interval excludes zero). Top block: primary rotation (leave-lab-out, 8 classes). Middle: test-time adaptation on C1. Bottom: second rotation (C1-s6out-aug paired against its own baseline) and the 5-class SHARP reproduction; neither is comparable to the top block.

Config	Objective	Val F1	Test F1	Test acc	$\Delta\text{acc vs C1}$ [95% CI]
C1	CE (our model)	0.887	0.804	0.805	—
C1' (s43)	CE, seed 43	0.878	0.799	0.800	-0.005
C2	CE + GRL	0.842	0.601	0.647	-0.158 [-0.323, +0.034]
C2' (s43)	CE + GRL, s43	0.787	0.562	0.600	-0.205
C3	SupCon (probe)	0.819	0.729	0.727	-0.078 [-0.128, -0.030]*
C3-ft	SupCon + CE ft	0.818	0.706	0.720	-0.085 [-0.221, +0.058]
C1-lin	C1 frozen + probe	0.884	0.806	0.807	+0.002
C1-aug	CE + amp. aug	0.883	0.772	0.776	-0.028 [-0.092, +0.023]
C1-sharplike	CE, SHARP net	0.738	0.543	0.569	-0.235 [-0.399, -0.092]*
C1+AdaBN	test-time BN	—	0.722	0.746	-0.059 [-0.193, +0.031]
C1+T3A	test-time protos	—	0.810	0.812	+0.007 [-0.047, +0.061]
C1+both	AdaBN + T3A	—	0.753	0.758	-0.047 [-0.142, +0.035]
C1-s6out	CE, living-out	0.776	0.684	0.743	(own rotation)
C1-s6out-aug	+ amp. aug	0.923	0.823	0.816	+0.073 [-0.029, +0.203]
C0	SHARP repro (P1, 5-cls)	0.892	0.605	0.612	(own split)

the negative result, test-time adaptation and augmentation, and finally the error structure and calibration of the deliverable model.

Statistical protocol. The held-out test sets are small (11 traces for the primary rotation), so we treat the *trace*, not the window, as the unit of statistical analysis. Differences between configurations are assessed with a trace-level *paired* bootstrap ($N = 10^4$ resamples, fixed seed): traces are resampled with replacement and the same resample is applied to both streams, so per-trace difficulty cancels. The bootstrap runs on *accuracy*: with six of eight classes represented by a single trace, only $\sim 2.8\%$ of trace resamples contain all eight classes, so a macro-F1 bootstrap would average a different class set in almost every replicate. Macro-F1 is therefore reported descriptively, without an interval. Two sources of uncertainty are kept separate: the bootstrap measures *test-sampling* variance, while *training* (seed) variance is measured by the seed-43 replicates. Finally, since ~ 10 comparisons share the same 11 test traces, we designate C1–C2 and C1–C3 — the two questions the study was designed to answer — as primary and treat the remaining comparisons as exploratory.

A. Main results

Tab. 2 reports validation and test performance for all configurations, with the paired accuracy difference against our model C1 and its 95% bootstrap interval. Three facts summarize it:

- **Our model transfers best.** C1 reaches macro-F1 0.804 on the held-out laboratory environment, no variant resolves an improvement over it, and it is seed-stable (its seed-43 replicate lands 0.5 accuracy points away).
- **Neither training objective helps.** SupCon and the GRL variant fall 7.8 and 15.8 accuracy points below it (Section VI-C).
- **The deep backbone matters most.** At equal recipe and class set, the shallow-wide SHARP network loses 23.5 points (Section VI-B).

B. Our model vs. the SHARP baseline

The faithful reproduction C0 confirms the reference implementation: trained on one bedroom capture set with the paper’s 5 classes and decision fusion, it reaches macro-F1 0.605 on the held-out capture sets — the same bedroom on later campaigns plus the two unseen rooms — with per-set accuracy between 0.50 and 0.73, consistent with the degradation reported in the original work. Because C0 runs on a different split and class set, the like-for-like architecture comparison is the ablation *C1-sharplike*: the SHARP network inserted into the exact C1 recipe (same split, classes, loss, optimizer, seed; only the backbone differs). It loses 23.5 accuracy points to C1 (test macro-F1 0.543 vs. 0.804), with an interval $[-0.399, -0.092]$ that excludes zero, after the same recipe had already shown a ~ 15 -point gap on the in-domain validation metric. The deep backbone is therefore decisively better both in domain and on the held-out environment — and since it was originally chosen on a throughput gate, the accuracy axis now vindicates that choice independently.

C. Do the training objectives help?

SupCon does not. C3 is below C1 under every readout we tried on its frozen validation features — linear probe 0.819, kNN 0.805, nearest-class-mean 0.718 (macro-F1), against C1’s 0.884, 0.856 and 0.889 respectively — so the frozen linear recipe is not what penalizes it: C1 is robust to the readout while C3 loses under all three. On test the paired difference -0.078 , CI $[-0.128, -0.030]$, resolves against it. Giving SupCon a “fair shot” by fine-tuning the whole network from its initialization (C3-ft) does not recover the gap either (0.706 on test, $\approx$ its own linear-probe ceiling): the SupCon representation is the limitation, not the frozen readout. A concatenation control on validation makes the point sharper: fusing $C1 \oplus C3$ features scores *below* a same-loss control $C1 \oplus C1'$ (0.868 vs. 0.888), so SupCon adds nothing that a second CE encoder would not.

The GRL costs, with high variance. On test C2 loses 15.8 accuracy points to C1. The direction is stable — it holds on both seeds (seed-43 gap: 20.0 points) and on validation — but the bootstrap interval $[-0.323, +0.034]$ crosses zero: with 11 test traces, even a 16-point gap is not resolved at the 95% level, and we report it as a direction with a range rather than a point estimate. The seed replicates add a second observation: the same-config seed spread is 0.5 accuracy points for C1 against 4.7 for C2 — with two seeds per configuration this is an observed range, not an estimate, but it suggests the adversary costs training stability as well as accuracy. A linear probe on C2’s *frozen features* reaches 0.720 accuracy on test, 7 points above C2’s own trained head (0.647): a substantial part of the GRL damage sits in the classifier head rather than in the representation. The damage is also concentrated, not uniform: at trace level C2 collapses on the H, W and S traces (on H, accuracy 0.04 vs. C1’s 0.64), matches C1 on the jumping and sitting traces, and is even ahead on one running trace.

TABLE 3: Domain readability of the C1 encoder: linear-probe accuracy vs. each target’s majority baseline, on a trace-disjoint inner split of train features. The activity control saturates; every domain attribute is at or below baseline (*: the probe converges to the constant majority predictor).

Target	Acc	Baseline	Δ
activity (control)	1.000	0.197	+0.803
capture set	0.196	0.286	−0.090
environment*	0.854	0.854	+0.000
direct path	0.633	0.731	−0.098
subject*	0.818	0.818	+0.000
monitor	0.499	0.584	−0.086

D. Why: the adversary has no target

The natural question is why an environment-invariance objective fails on an environment-transfer benchmark. We answer it by probing the *frozen* encoders directly. On a trace-disjoint inner split of the training features we train linear probes for the activity (positive control) and for every domain attribute, each against its own majority baseline (Tab. 3). The activity control saturates, while *every* domain attribute sits at or below its majority baseline: the environment, subject and hardware attributes are not linearly readable from the CE features, on precisely the data — train features — that an in-training adversary gets to read. The same probes on the C2 (GRL) and C3 (SupCon) encoders, and on a second rotation whose train environments differ, give the same verdict. Fig. 1 shows the geometric counterpart: activities form tight clusters, while capture sets are uniformly mixed within every cluster.

The implication is structural. A GRL removes domain information that the discriminator can read; here there is nothing to read, so the adversary can only inject gradient noise — which is exactly what the training monitors showed (the domain head never rose meaningfully above the majority-class floor in any GRL run, on either seed) and what the results confirm (a cost with no invariance gain). The root cause is the composition of the training split: one environment dominates it, and the second is represented by a single capture set, so environment-invariance is barely definable on this training distribution. On this evidence C4 (SupCon+GRL) was closed without training: with no readable domain under either loss family, its expected outcome is “C3 plus noise”. We report this as an evidence-based triage decision.

E. Test-time adaptation and augmentation

TTA is neutral at best. T3A is statistically indistinguishable from plain C1 (+0.007, CI [−0.047, +0.061]); AdaBN is negative (−0.059, CI [−0.193, +0.031]) and drags the combination down with it (−0.047). Re-estimating BN statistics on a single-environment, 11-trace unlabeled test set moves the statistics away from a training distribution that, per the diagnostics above, was not environment-specific to begin with — consistent with our central finding.

Amplitude augmentation is a regularizer, not a cross-domain lever. The pre-registered discriminator was: if the

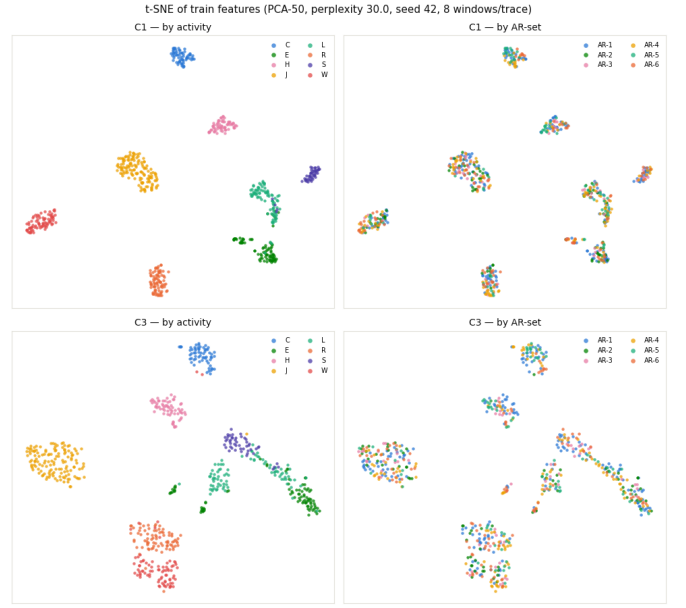


Fig. 1: t-SNE [9] of train features. Top row: C1 (CE); bottom row: C3 (SupCon). Left column: colored by activity (tight clusters form; in C3 the L/S/E clusters chain into one filament). Right column: the same points colored by capture set — uniformly mixed within every cluster: no domain substructure exists for an adversary to remove.

augmentation acts through cross-domain invariance, its test gain should exceed its validation gain. The opposite happened. On the primary rotation, where the baseline was already near its ceiling, the augmentation changed nothing on validation (−0.004) and slightly hurt on test (−0.028, CI crossing zero). On the living-room-out rotation, whose baseline is much lower, it helped on validation (+0.147) and on test by a comparable amount (+0.073, CI [−0.029, +0.203]): a gain where there was headroom, of the same order in and out of domain — the signature of generic regularization on a small training set.

F. Error structure and calibration of the deliverable model

Per-class behaviour refutes a declared caveat. The validation split of the primary rotation contains no C, E or S traces, so checkpoint selection was blind to three classes and we pre-declared the risk that they would underperform on test. The opposite is true: the val-blind classes are the *best* (mean F1 0.904: C 0.933, E 1.000, S 0.780) while the val-seen ones average 0.744. On the second rotation the picture flips — a val-seen class collapses (S, F1 0.131) while the blind classes sit mid-pack. Selection blindness is therefore not the driver of per-class error on these test sets; class difficulty in the target environment is.

Errors are physically sensible. C1’s dominant confusion on the held-out environment is walking↔running (18 windows in each direction — adjacent Doppler signatures), followed by H→J (14, two whole-body dynamic classes), while the remaining mass falls among the low-motion classes

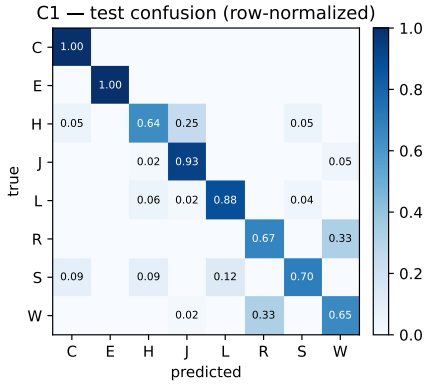


Fig. 2: Confusion matrix of our model C1 on the held-out laboratory environment (row-normalized). The dominant confusions are walking $\leftrightarrow$ running and H $\rightarrow$ J.

(S is spread over L, H and C); the empty-room class is recognized perfectly. The confusion matrix is shown in Fig. 2. Errors between physically similar activities, rather than scattered ones, indicate that the model transfers genuine motion features.

Calibration. C1 is systematically *underconfident* on test (ECE 0.243; accuracy above confidence in every reliability bin), an expected effect of the label smoothing in its loss. The GRL model is the best calibrated (ECE 0.070) while being the least accurate — calibration and accuracy dissociate across the loss families — and T3A, which sharpens predictions, worsens calibration (ECE 0.345). Any deployment that consumes confidence scores therefore needs a temperature-scaling stage, whatever the configuration.

VII. CONCLUDING REMARKS

In this project we evaluated, under one shared apparatus and a single-use test protocol, whether adversarial (GRL) and contrastive (SupCon) training objectives improve cross-environment WiFi CSI activity recognition over a plain cross-entropy model. They do not: SupCon is worse by a bootstrap-resolved margin; the GRL is worse still, with a much larger seed-to-seed spread; and test-time adaptation is neutral at best. The deep CE model itself, however, outperforms the SHARP-style shallow network by 23.5 accuracy points on the unseen environment — on this benchmark, backbone capacity mattered far more than any invariance objective.

The practical observation we consider most useful is *why* the objectives fail: frozen-encoder probes show that no domain attribute is linearly readable from the learned features, so a domain adversary has nothing to remove. This precondition is cheap to measure — a linear probe on cached features — and we suggest checking it before investing in adversarial or contrastive invariance on any dataset whose training split, like this one, is dominated by a single environment. Benchmark designers, conversely, need several well-populated training environments for domain-generalization claims to be meaningful.

A. Challenges and Lessons Learned

Most of our effort went into making a negative result trustworthy: splits frozen on day one and split by trace, normalization computed on train only, a single logged test session with validation-selected checkpoints, and a pre-registered row list. We learned to distrust single numbers twice. First, the seed replicates showed a same-configuration swing on the GRL model larger than some cross-configuration gaps, forcing us to report directions with ranges rather than point estimates. Second, with 11 test traces even a 16-point accuracy gap is not resolved at the 95% level by a trace-level bootstrap — the honest unit of analysis (the trace, not the thousands of correlated windows) makes the dataset look much smaller than it appears. A declared caveat also failed empirically: the classes invisible to validation turned out to be the *best* on test, a reminder that plausible methodological worries should be tested, not just declared.

B. Future Work

With one held-out environment per rotation, this is a transfer study rather than full domain generalization; the natural extension is a dataset with several well-populated training environments and a complete leave-one-out over all of them, where our diagnostic protocol transfers unchanged. On the modeling side, the open question is whether *any* training-time objective can help when the domain factor is unreadable — candidates that operate on the input rather than the representation (stronger physics-informed augmentations, synthetic environment simulation) are unaffected by our structural argument.

REFERENCES

- [1] Z. Wang, K. Jiang, Y. Hou, W. Dou, C. Zhang, Z. Huang, and Y. Guo, “A Survey on Human Behavior Recognition Using Channel State Information,” *IEEE Access*, vol. 7, pp. 155986–156024, 2019.
- [2] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, “Domain-Adversarial Training of Neural Networks,” *Journal of Machine Learning Research*, vol. 17, no. 59, pp. 1–35, 2016.
- [3] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, “Supervised Contrastive Learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, (Online), Dec. 2020.
- [4] F. Meneghello, D. Garlisi, N. D. Fabbro, I. Tinirello, and M. Rossi, “SHARP: Environment and Person Independent Activity Recognition with Commodity IEEE 802.11 Access Points,” *IEEE Transactions on Mobile Computing*, vol. 22, pp. 6160–6175, Oct. 2023.
- [5] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A Simple Framework for Contrastive Learning of Visual Representations,” in *International Conference on Machine Learning (ICML)*, (Online), July 2020.
- [6] Y. Li, N. Wang, J. Shi, J. Liu, and X. Hou, “Revisiting Batch Normalization for Practical Domain Adaptation,” in *International Conference on Learning Representations (ICLR) Workshop*, (Toulon, France), Apr. 2017.
- [7] Y. Iwasawa and Y. Matsuo, “Test-Time Classifier Adjustment Module for Model-Agnostic Domain Generalization,” in *Advances in Neural Information Processing Systems (NeurIPS)*, (Online), Dec. 2021.
- [8] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, (Las Vegas, NV, US), June 2016.
- [9] L. van der Maaten and G. Hinton, “Visualizing Data using t-SNE,” *Journal of Machine Learning Research*, vol. 9, no. 86, pp. 2579–2605, 2008.